



Pioneering Financial Data Extraction and Aggregation

How Envestnet | Yodlee retrieves and aggregates financial data to power financial innovation.

Innovation in digital financial services requires the very best in data and data intelligence. Envestnet® | Yodlee® provides both. As the pioneer in data aggregation, Envestnet | Yodlee delivers significant depth and breadth of financial data by aggregating from more than 15,000 data sources—including more than 1,000 companies with more than 22 million registered users and hundreds of millions of transactions. The data network is sustained through close partnerships with major financial institutions, with over 71 percent of the data collected through direct feeds.

By leveraging this rich data, financial institutions and fintech providers can uncover new opportunities to serve customers, deliver actionable guidance for consumer financial applications, and offer a truly personalized and seamless user experience.

Data Gathering Methods

Envestnet | Yodlee aggregates content on a daily basis from over 15,000 connections from financial institutions, billers, reward networks, and other end points on behalf of the consumer. The user-permissioned data is gathered from these end points either through a direct feed with financial institution partners, screen scraping, or a combination of these methods.

Data Feeds

Envestnet | Yodlee technology is capable of obtaining consumer information from most partner websites in a variety of ways, the preferred method of obtaining this information is through a direct data feed. Using a direct data feed, our partners can offer the most recent data and a higher quality experience to the consumer base that utilizes sites and apps powered by the Envestnet | Yodlee platform.

The benefits of a direct data feed are numerous:

- Users enjoy a quick response time when retrieving and updating information
- Information gathered by Envestnet | Yodlee is not affected when there's a change to the content website
- Information is directly shared by the financial institution, resulting in a lift in data quality
- There is a reduced load on a financial institution's website since Envestnet | Yodlee does not have to log in and navigate to multiple pages on the financial institution's website

In order to offer this higher quality of service, our content provider customers give Envestnet | Yodlee with access to a Data Feed Server (DFS) that can be queried for information. The data is aggregated through the data feed, resulting in a seamless user experience.

Data Feed Technical Overview

Envestnet | Yodlee information systems are built on industry standard data formats and communication protocols such as OFX, XML, HTTP, SSL, and TCP/IP. XML is an extensible markup language that is a worldwide standard specified by the World Wide Web Consortium (W3C). HTTP, SSL, TCP, and IP are the application, security, transport, and network level protocols specified by the Internet Engineering Task Force (IETF) that web clients and web servers use to communicate on the Internet.

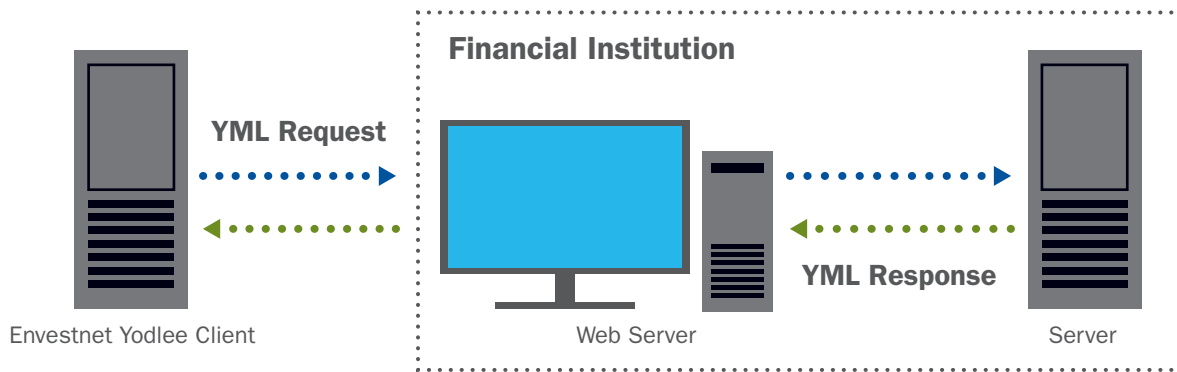
We recommend 2nd generation OFX with some additional Envestnet | Yodlee specific data nodes. Envestnet | Yodlee defines a data feed request/response mechanism and also the data model specific to a container (i.e., Card, Bank, Loan, Investments). The data feed software sends the request in the specified format, which prompts a response from the data feed server, also in the same format.

A container defines a particular type of aggregated data. Accounts of the same type are tagged together that belong to a single container. In cases where Envestnet | Yodlee has to pull data, we make container-specific calls to extract information for a set of accounts in each container for an individual user. For example: Banking, Credit Card, Loan, and Investments.

Request/Response Model

The Envestnet | Yodlee data model is currently designed for a single user. This means we send individual requests for every consumer and receive the responses for each of them.

Envestnet | Yodlee requests data from content partners, preferably in the OFX format. This request is called the request block. After establishing connection to a content partner's Data Feed Server using SSL, Envestnet | Yodlee sends an HTTP Post request with the request block. The request block contains the username, password, and other required parameters.



NOTE: Typically the standard for data feed is OFX. However, if you have an IFX or XML based data feed, Envestnet | Yodlee will leverage it and map it to the data model. Envestnet | Yodlee creates the OFX request having SIGNONMSGSRQ and SIGNUPMSGSRQ messages and sends it to the partner data feed server. The partner data feed server sends the response back with all the accounts for that consumer to Envestnet | Yodlee. We extract the account for the specific container for which the consumer has aggregated. Then Envestnet | Yodlee creates another OFX request having SIGNONMSGSRQ and a respective request for getting container details and transactions.

Screen Scraping

Since 1999, Envestnet | Yodlee has been working with our financial institution partners to deliver the most secure and private data aggregation service platform available. This allows consumers secure and unfettered access to their data through bank and third-party non-banking sites so they can utilize it for the tools and services designed to improve their financial lives. Envestnet | Yodlee has been a leader in this area and has supported a variety of new and ongoing initiatives to this end, such as direct feed API, Enterprise API, and Open Authentication (OAuth), a token-based security authentication technology.

All types of access are supported by Envestnet | Yodlee, and we have continually encourage our financial institution partners to support the latest and greatest security and technology solutions such as direct feed API and OAuth security.

For aggregation technology, Envestnet | Yodlee prefers and supports direct data feeds and we are diligently working with many of our financial institutions, and other partners to enable that. Envestnet | Yodlee is leading the way in API integration with over 71 percent of our data in network coming through direct feeds. We firmly believe broader adoption of this model will benefit everyone in terms of addressing technical requirements, efficiency, and security as well as user needs and experience.

In those situations where we do support screen scraping, Envestnet | Yodlee has painstakingly built this connectivity working with our banking partners to enable a high quality service that effectively conforms to the needs and

requirements set by our banking partners. This type of relationship is critical in these situations to insure our efforts effectively reduce the frequency of traffic-inducing data requests hitting a financial institutions infrastructure and significantly impact their responsiveness and customer service queues.

Some of the benefits of screen scraping include the following:

- Agents are capable of navigating to different web pages and retrieving data to be displayed to the consumer on the partner's website, at any point of time.
- Agents can be enhanced by Envestnet | Yodlee directly to scrape additional information whenever needed.
- As it is difficult to establish a partnership with each and every financial institution for data feeds, scraping can be an ideal option for gathering information from such sites.
- Screen scraping can be a cost-effective method of retrieving data, especially for smaller partner sites with less data retrieval requirements.

While the screen scraping method of data retrieval is fairly robust, there are a few considerations:

- It takes several seconds, sometimes up to 30 seconds or more to retrieve data, especially the first time an account is added. Envestnet | Yodlee systems may have to parse through several web pages in order to retrieve the necessary information, potentially while the consumer is waiting.
- Not all data in the Envestnet | Yodlee data model is retrievable from each website. Occasionally websites will not make certain data points available to their users. As an example, some credit card providers don't have the APR for a consumer's credit card on the site.
- At times, the partner's website may be unable to handle high volumes of requests, which may lead to sub-optimal response times for the end user.
- Periodic maintenance may be required by Envestnet | Yodlee to account for changes at the partner's website. If a partner drastically changes web page formats without notice, the agent may not be able to pull the data correctly.

Summary Account Types Supported

	Bank	Credit Card	Investment	Loan	Insurance	Reward
Account	• Institution Name • Account Name • Account Type • Balance • Rate	• Institution Name • Account Name • Statement • Balance • Rate	• Institution Name • Account Name • Account Type • Statement • Holdings	• Institution Name • Account Name • Account Type • Statement • Account • Balance • Rate	• Institution Name • Account Name • Premium amount • Balance • Statement	• Program Name • Program Type • Balance
Transactions	Transaction – Date, Amount, Merchant Name, Category, Merchant Geo (Street, City, State, Zip, Latitude, Longitude), Status, Currency, Credit/Debits	Transaction – Date, Amount, Merchant Name, Category, Merchant Geo (Street, City, State, Zip, Latitude, Longitude), Status, Currency, Credit/Debits	Transaction – Date & Type, Amount, Holding Name & Type, Quantity, Price, Cost Basis, CUSIP Symbol, etc.	Transaction – Date, Amount, Merchant Name, Payments	Transaction – Date, Amount, Merchant Name, Payments	Transaction – Date, Description, Amount

Envestnet | Yodlee's customer collaboration model and data gathering strategy means we partner with our banks in these efforts by working in lock step with our financial institution partners to deliver valuable services to their end users without creating any undue burden for the financial institution or their customers.

Types of Data Gathered

Envestnet | Yodlee uniquely and securely obtains user-permissioned data using direct feeds and patented aggregation technology to provide the needed bank, credit card, investment, loans, insurance rewards, and other financial account data.

Data Enrichment

Once the data has been brought in from the end point, Envestnet | Yodlee cleanses and augments the data to ensure that the data is consistent and usable through Envestnet | Yodlee Transaction Data Enrichment. This

solution takes confusing, jumbled transaction data and translates it into clear, easy-to-read information, with a merchant name, date, amount, and a simple description of the transaction. Envestnet | Yodlee also offers categorization and analytics in addition to basic cleansing, which allows partners to obtain deeper insights and create more advanced user experiences.

Whichever data retrieval method you choose, Envestnet | Yodlee makes it possible to obtain the best quality data available at fast speeds. The resulting data intelligence can be used to personalize your apps, services, and advice, fully engage your users, and completely revolutionize the customer experience.



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